More direct searches for the dark matter particles streaming through the earth have also been under way for some time, although these are extremely challenging experiments. Of the million or so dark matter particles that should be passing through an area the size of a quarter each second, at most one per day would leave any evidence in the specially designed equipment that various experimenters have built to detect them. To date, no confirmed detection of a dark matter particle has been achieved.[10] With the prize still very much up in the air, researchers are pressing ahead with much intensity. It is quite possible that within the next few years, the identity of the dark matter will be settled.

Definitive confirmation that dark matter exists, and direct determination of its composition, would be a major advance. For the first time in history, we would learn something that is at once thoroughly basic and surprisingly elusive: the makeup of the vast majority of the universe's material content.

All the same, as we saw in Chapter 10, recent data suggest strongly that even with the identification of the dark matter, there would still be a significant plot twist in need of experimental vetting: the supernova observations that give evidence of an outward-pushing cosmological constant accounting for 70 percent of the total energy in the universe. As the most exciting and unexpected discovery of the last decade, the evidence for a cosmological constant—an energy that suffuses space—needs vigorous, airtight confirmation. A number of approaches are planned or already under way.

The microwave background experiments play an important role here as well. The size of the splotches in Figure 14.4—where, again, each splotch is a region of uniform temperature—reflects the overall shape of the spatial fabric. If space were shaped like a sphere, as in Figure 8.6a, the outward bloating would cause the splotches to be a bit bigger than they are in Figure 14.4b; if space were shaped like a saddle, as in Figure 8.6c, the inward shrinking would cause the splotches to be a bit smaller; and if space were flat, as in Figure 8.6b, the splotch size would be in between. The precision measurements initiated by COBE and since bettered by WMAP strongly support the proposition that space is *flat.* Not only does this match the theoretical expectations coming from inflationary models, but it also jibes perfectly with the supernova results.